# Syllabus ESM 262 - Spring 2026

## A bit about me

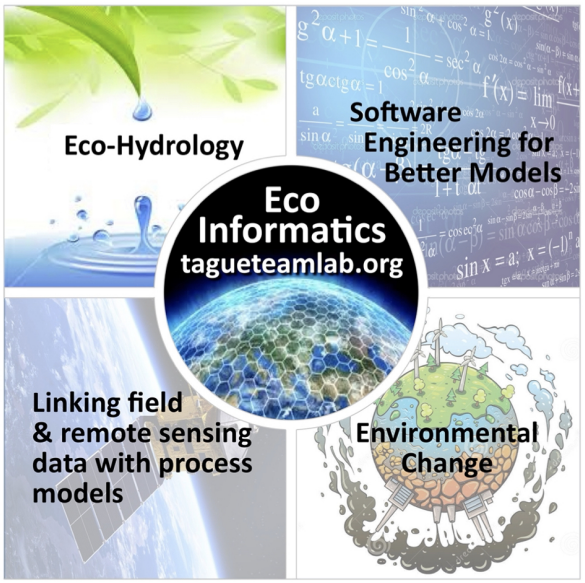


Figure 1: Growing skills

## Course description

ESM 262 is an introduction to computing for environmental applications. The course provides practical training in software design best practices.

- programming language concepts
- loops and repetition
- data types and structures

- modular program design;
- reproducibility and version control;
- testing, documentation, and workflow;

The course features **R** for programming, **Git** for version control, **Quarto** for workflow and editing, and **GitHub** for collaboration and publishing, but many concepts would be applicable in other software design tools.

Class will include a mix of lectures and hands-on examples, using students' own computers.

## Course where and when

**T/TH** 8:00 AM - 9:15 AM (Bren Hall 1424)

Teaching Team

**Instructor:** Naomi Tague (www.tagueteamlab.org)

- **Office hours:** email me
- **email** (tague@ucsb.edu)
- **git user id** naomitague

**Teaching assistant:** Ojas Sarup

- **Office hours:** Thursday 11am - 1pm (Bren Hall 1005)
- **email** (ojassarup@ucsb.edu )
- **git user id** osarup

## Learning objectives

- practice partner coding and reproducible workflows
- 3 *BIG* concepts in programming are *modularity*, *data structure*, *repetition* - we will learn skills related to all three of these
- learn and practice some coding/programming best practices (that will make your data science life easier - whatever you do!)
  - documentation
  - testing
  - reproducibity
- learn some new Rskills that are helpful for a wide variety of data science applications

### Computing set up

- Please complete the install-guide on the course website
- Please bring your laptop to class

### Course materials and how we will work

All lecture materials will be available on the course website as well as assignments. But you will submit assignments on Canvas

https://naomitague.github.io/ESM_262/

Class time will involve lectures and working on practical applications . This way we use the class time as a lab - where you can get guidance.

Why structure it this way? You learn to code by doing - not so much by watching others do!

## Using AI-tools in this course

- AI-tools great resources but effective use requires understanding how to ask good questions
- Use this course to build that understanding, think about the meta concepts.
- Ask yourself - What are situations you might apply a technique/idea/concept;
- You gain skills by **Practicing**, so start without AI-tools
- For assignments/practice
  - **Always** write the code yourself -
  - Supportive ways to use AI-tools
    - use AI-tools to help find syntax
    - Use AI-tools to explain other peoples code (including class examples)
    - Use AI-tools to help interpret error messages

## Attendance

- You are expected to come to class - its part of the learning!
- We will regularly have in-class quizzes - These are not really "quizzes" where you answer is right or wrong. They are designed to help you think through the material and provide feedback to me as the instructor. For each quiz you will get full credit for attempting the question in class.
- If you have to miss class let Ojas and me know by email

### Tentative Schedule

| # | Date | Type | Topic |
|---|---|---|---|
| 1 | Tue, Apr 14 | Regular | Course overview and setup |
| 2 | Thu, Apr 16 | Regular | Quarto |
| 3 | Tue, Apr 21 | Regular | Git and GitHub |
| 4 | Thu, Apr 23 | Regular | Git and Collaboration |
| 5 | Tue, Apr 28 | Regular | Git a bit more |
| 6 | Thu, Apr 30 | Regular | Data Types and Structures, Lists |
| 7 | Tue, May 5 | Regular | Looping |
| 8 | Thu, May 7 | Regular | Modularity |
| 9 | Mon, May 11 | Evening | Functions! |
| 10 | Tue, May 12 | Regular | Functions and Data |
| 11 | Thu, May 14 | Regular | Flow Control |
| 12 | Mon, May 18 | Evening | Project Design, Modularity and Workflow - Practicing |
| 13 | Tue, May 19 | Regular | APIs and servers |
| 14 | Thu, May 21 | Regular | Working with the shell |
| 15 | Tue, May 26 | Regular | Testing and debugging |
| 16 | Thu, May 28 | Regular | Documentation |
| 17 | Tue, Jun 2 | Regular | Putting it together |
| 18 | Thu, Jun 4 | Regular | Where does AI fit |

### Assignments

There will be 6 assignments (more or less one for each week). You will usually have time to start the assignment in class and most will be in groups. We will also have 6 quizes

## Assignment expectations and grading

- All assignments submitted on Canvas
- Late Assignments 10% each day. Try to submit on time as we build on the concepts and materials covered in each assignment
- Style counts; So make sure your follow good programming practices including adding documentation and using informative variable names
- You can use AI (ChatGPT etc) to help you learn syntax, but write the code yourself; this is how you learn
- You can resubmit your assignment for additional feedback and we will upgrade your grade. Please do so with in one week of getting the assignment back